

Ameya Kshirsagar

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(Open to Relocate)

SUMMARY

Result Oriented **Data Analyst** with 3+ years of experience in education, energy, and healthcare. Expert in **Python, SQL, Machine Learning**, and **Data Visualization (Tableau, Google Looker)**. Delivered data-driven solutions that increased user engagement by 50% improved feature accuracy by 40%, and enhanced reporting efficiency by 30%. Skilled in predictive modeling, feature engineering, **A/B testing**, and statistical analysis. Proven ability to optimize data workflows and provide actionable insights. **Actively seeking opportunities to leverage** expertise and contribute to organizational success in a challenging role

RESEARCH, AWARDS, RECOGNITION, CERTIFICATIONS

- Only an Indian candidate was selected to represent at the Stanford Geothermal Conference. Authored a research paper on geothermal energy, showcasing the application of machine learning in identifying and classifying geothermal micro-seismic events

Authored two books (Data Science domain)

- Book 1 - Transforming Agriculture Technology Using AI and Robotics Available for purchase on Amazon
- Book 2 - Applications of Artificial Intelligence (AI) and Machine Learning (ML) in the Petroleum Industry Available for purchase on Amazon
- Presented research and insights on AI and Data Science topics (seismic activity) at Stanford University.
- Active Research Reviewer with Elsevier and Springer
- Certified Machine Learning Specialist.

PROFESSIONAL EXPERIENCE

FIELD DAY LEARNING GAMES

Remote, US

Data Analyst

Jun 2023 - Present

Led data initiatives for educational games to enhance learning engagement, focusing on feature engineering, data processing, data analysis, statistical modeling, predictive modeling, and delivering actionable insights to stakeholders.

Game Name: BLOOM: Environmental Policy Simulation game

- Designed and implemented **event-driven analytics** using **SQL** and **Python**, including features like **BuildCount** and **PolicyChangeCount**, improving tracking accuracy and system efficiency
- Developed and deployed **Machine Learning** models to analyze player behavior, optimize game mechanics, and deliver predictive insights, driving a 50% increase in user engagement and enhancing educational simulation outcomes
- Utilized **Python (NumPy, Pandas, Matplotlib, Scikit-learn)** and **R** for predictive analytics, statistical modeling, and deep learning techniques to analyze player behavior, optimize gameplay mechanics, and deliver actionable insights that enhanced user engagement and learning outcomes across multiple educational simulation projects
- Extracted, transformed, and analyzed large datasets using **SQL** across **PostgreSQL, MySQL, and MSSQL**, optimizing query performance and generating actionable insights for data-driven decision-making
- Designed and maintained **ETL** pipelines to automate data collection, transformation, and storage, improving data accessibility for analytics teams
- Leveraged **Microsoft Azure** services to analyze and visualize large datasets, utilizing cloud-based storage and computing solutions to drive data-driven insights and reporting efficiency
- Developed interactive dashboards in **Tableau, Power BI and Looker**, visualizing player behavior and performance metrics, enhancing stakeholder insights, and driving a 50% improvement in user engagement
- Experienced in utilizing **Excel's** advanced functionalities, including **Pivot Tables, VLOOKUP, Macros, and VBA**, to analyze and summarize gameplay data, streamline reporting processes, and enhance stakeholder reporting accuracy and efficiency across multiple educational game projects
- Created automated **KPI dashboards** to track business performance, user retention, and feature adoption, ensuring data-driven strategy optimization
- Utilized **JIRA** and **Agile methodologies** to track project progress, manage tasks, and collaborate with cross-functional teams, ensuring seamless execution of data workflows and timely delivery
- Delivered granular insights with subfeatures and dashboards, increasing feature completeness by 40% and aiding stakeholder reporting

Game Name: ThermoLab & ThermoVR: VR-based Interactive thermal science education game

- Created detectors like SliderMoveDetector to track player interactions in VR simulations, optimizing gameplay data accuracy

- Developed **Python**-based event-driven tracking systems and analytical tools to process gameplay data, enabling enhanced feature engineering, predictive modeling, and actionable insights, which improved user engagement by 50% across multiple educational simulations
- Performed **A/B testing** and statistical analysis to evaluate the effectiveness of gameplay features and optimize player retention strategies
- Proficient in **SQL Server** with expertise in creating and utilizing stored procedures, user-defined functions, sub-queries, joins, and exception handlers for complex multi-table queries
- Improved user interaction tracking by 30%, enhancing gameplay refinement and learning outcomes

Game Name: AquaLab: Ecosystem modeling and restoration game

- Developed event-driven tracking systems using **Python** and **Pandas** to analyze player decisions in an environmental simulation
- Contributed insights that improved feature clarity, boosting player retention and engagement by 25%

General Contributions

- Built scalable data analysis solutions using **Python**, **SQL**, and **Tableau**, enhancing game mechanics and user insights
- Analyze player behavior and develop data-driven solutions to optimize game design, enhance user engagement, and provide actionable insights for educational simulations.
- Increased user engagement by 50% and improved stakeholder reporting across multiple simulation projects

PANDIT DEENDAYAL ENERGY UNIVERSITY (PDPU)

Gujarat, India

Data Science Research Executive

May 2020 - Aug 2022

Applied and conducted advanced research in data science, collaborating with IISc Bangalore (India's #1 institute) on photovoltaic optimization while developing machine learning models for renewable energy, healthcare, education, financial, and petroleum industries.

Client /Collaboration: IISc Bangalore

Photovoltaic optimization for renewable energy in 3D photovoltaic cells.

- Collaborated on a project to optimize prediction accuracy to 96.8% by analyzing organic 3D photovoltaic cell data, improving key metrics such as JSC (Short-Circuit Current Density), VOC (Open-Circuit Voltage), FF (Fill Factor), and Efficiency (%).
- Leveraged Python, and machine learning techniques to analyze and deconstruct unit data, contributing to advancements in renewable energy research.
- Presented findings to a cross-functional research team, enhancing the scientific understanding of photovoltaic performance optimization.

Internal Projects

- Integrated machine learning applications in the healthcare and petroleum industries, forecasting COVID-19 trend patterns and advancing predictive capabilities.
- Co-authored a study on machine learning applications in stock market prediction, published in Decision Analytics Journal (Elsevier, 2021), exploring artificial neural networks, support vector machines, and short-term memory models.
- Led and mentored a team of 2+ research associates to deliver high-quality insights for academic and industrial projects.
- Authored two books and published multiple research papers while actively reviewing submissions for international AI and data science journals.

Client: Disha Consultant

The client aimed to optimize their recruitment process by analyzing historical hiring data to identify patterns, improve candidate selection efficiency, and enhance overall recruitment strategies through data-driven insights.

- Designed analytical reports to assess education and career trends, enabling better career guidance services.
- Created dynamic dashboards to visualize employment and higher education metrics, providing actionable insights for decision-making.
- Improved data reporting and analytics enhanced client satisfaction and engagement rates by 20%.

OMDENA

New York, US

Machine Learning Engineer

Aug 2021 – Nov 2021

Client: Transformative.ai

The client aimed to develop a Cardiac Arrest Prediction System by leveraging comprehensive datasets collected from multiple hospitals to enhance early detection and intervention strategies.

- Developed machine learning models for cardiac arrest prediction, achieving an accuracy of 97.05% and an area under the curve (AUC) of 0.91.
- Predicted cardiac arrest caused by pulseless electrical activity and asystole, contributing to the design of innovative healthcare solutions.
- Optimized backend performance, enhancing system efficiency and scalability for real-time healthcare applications.
- Led a team of 11 members to accomplish sub-tasks and ensure timely project delivery.

Client: INACTIVE Cairo:

The client sought to harness advanced machine learning techniques to analyze satellite imagery and socioeconomic data, enabling the development of actionable solutions for urban planning and disaster management in Egypt.

- Utilized AI techniques and satellite-image analysis to address Egypt's critical water scarcity issues, focusing on reducing freshwater wastage from flood irrigation (77% of Nile water usage).
- Conducted data collection and visualized data to study water consumption habits and irrigation practices across Egypt.
- Predicted a reduction in freshwater wastage by approximately 11%, supporting sustainable resource management amidst regional water security challenges.
- Collaborated with a multidisciplinary team to propose innovative solutions for integrated water management and decision-making.

PERSISTENT SYSTEMS LTD.

Pune, India

Machine Learning Intern (NLP)

Jan 2019 – Jul 2019

The client sought to automate data entry for thousands of handwritten forms, aiming to significantly reduce manual effort by implementing an intelligent system to auto-populate data from scanned documents.

- Automated the insurance claims process by implementing advanced image pre-processing techniques and Optical Character Recognition (OCR) using PyTesseract, reducing manual intervention and processing time by 40%.
- Developed end-to-end solutions for claim reimbursement systems by leveraging Natural Language Processing (NLP) for entity extraction, improving data accuracy and efficiency in extracting key policyholder information.

PROJECTS

Data Science: Jobs, Salary, Skills, and Education

- Conducted quantitative research on data scientist salary trends and curriculum, bridging wage gaps and improving recruiting diversity.
- Achieved a 35% increase in diverse recruitment through actionable recommendations for education and industry stakeholders.

Improving University's Interlibrary Loan (ILL) Request System (2022)

- Identified inefficiencies in the university's ILL Request System, analyzing pain points with stakeholders.
- Developed a risk management plan, reducing project risks by 65% and enhancing system performance.

Skills

Data Visualization: Tableau, Colab, Excel, Google Looker

Databases: MySQL, NoSQL

Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), C++

Tools & Libraries: Excel, PowerPoint

Machine Learning: Supervised ML (Linear and Logistic Regression, Decision Trees, RF, Xgboost), Unsupervised ML (K-means, PCA), NLP, Classification techniques (binary classification, multi-class classification), A/B testing

Process Optimization: Lean Six Sigma

Strong analytical skills with experience in building scalable solutions and dashboards

EDUCATION

UNIVERSITY OF WISCONSIN-MADISON

Wisconsin, US

MS in Information Science (Data Science) (GPA: 3.8/4)

Sep 2022 - May 2024

SYMBIOSIS INTERNATIONAL UNIVERSITY

Maharashtra, India

BTech. in Information Technology

July 2016 - Apr 2020